Second, it may well be that processes of input analysis are fast *because* they are mandatory. Because these processes are automatic, you save computation (hence time) that would otherwise have to be devoted to deciding whether, and how, they ought to be performed. Compare: eyeblink is a fast response *because* it is a reflex—i.e., because you don't have to *decide* whether to blink your eye when someone jabs a finger at it. Automatic responses are, in a certain sense, deeply unintelligent; of the whole range of computational (and, eventually, behavioral) options available to the organism, only a stereotyped subset is brought into play. But what you save by indulging in this sort of stupidity is *not having to make up your mind,* and making your mind up takes time. Reflexes, whatever their limitations, are not in jeopardy of being sicklied o'er with the pale cast of thought. Nor are input processes, according to the present analysis.

There is, however, more than this to be said about the speed of input processes. We'll return to the matter shortly.

III.5. Input systems are informationally encapsulated

Some of the claims that I'm now about to make are in dispute among psychologists, but I shall make them anyway because I think that they are true. I shall run the discussion in this section largely in terms of language, though, as usual, it is intended that the morals should hold for input systems at large.

I remarked above that, almost certainly, understanding an utterance involves establishing its analysis at several different levels of representation: phonetic, phonological, lexical, syntactic, and so forth. Now, in principle, information about the probable structure of the stimulus at any of these levels could be brought to bear upon the recovery of its analysis at any of the others. Indeed, in principle *any* information available to the hearer, including meteorological information, astrological information, or—rather more plausibly—information about the speaker's probable communicative intentions could be brought to bear at any point in the comprehension process. In particular, it is entirely possible that, in the course of computing a structural description, information that is specified only at relatively high levels of representation should be 'fed back' to determine analyses at relatively lower levels.[20] But though this is possible in principle, the burden of my argument is going to be that the operations of input systems are in certain respects unaffected by such feedback.

I want to emphasize the 'in certain respects'. For there exist, in the psychological literature, dramatic illustrations of the effects of information feedback upon some input operations. Consider, for example, the 'phoneme restoration effect' (Warren, 1970). You make a tape recording of a word (as it might be, the word "legislature") and you splice out one of the speech sounds (as it might be, the 's'), which you then replace with a tape recording of a cough. The acoustic structure of the resultant signal is thus /legi(cough)lature/. But what a subject will *hear* when you play the tape to him is an utterance of /legislature/ with a cough 'in the background'. It surely seems that what is going on here is that the perceived phonetic constituency of the utterance is determined not just by the transduced information (not just by information specified at *sub*-phonetic levels of analysis) but also by higher-level information about the probable lexical representation of the utterance (i.e., by the subject's guess that the intended utterance was probably /legislature/).

It is not difficult to imagine how this sort of feedback might be achieved. Perhaps, when the stimulus is noisy, the subject's mental lexicon is searched for a 'best match' to however much of the phonetic content of the utterance has been securely identified. In effect, the lexicon is queried by the instruction 'Find an entry some ten phones long, of which the initial phone sequence is /legi/ and the terminal sequence is /lature/.' The reply to this query constitutes the lexical analysis under which the input is heard.

Apparently rather similar phenomena occur in the case of visual scotoma (where neurological disorders produce a 'hole' in the subject's visual field). The evidence is that scotoma can mask quite a lot of the visual input without creating a phenomenal blind spot for the subject. What happens is presumably that information about higher-level redundancies is fed back to 'fill in' the missing sensory information. Some such process also presumably accounts for one's inability to 'see' one's retinal blind spot.

These sorts of considerations have led to some psychologists (and many theorists in AI) to propose relentlessly top-down models of input analysis, in which the perceptual encoding of a stimulus is determined largely by the subject's (conscious or unconscious) beliefs and expectations, and hardly at all by the stimulus information that transducers provide. Extreme examples of such feedback-oriented approaches can be found in Schank's account of language comprehension, in Neisser's early theorizing about vision, and in 'analysis by synthesis' approaches to sentence parsing. Indeed, a sentimental attachment to what are known generically as 'New Look' accounts of perception (Bruner, 1973) is pervasive in the cognitive science community. It will, however, be a main moral of this discussion that the involvement of certain sorts of feedback in the operation of input systems would be incompatible with their modularity, at least as I propose to construe the modularity thesis. One or other of these doctrines will have to go.

In the long run, which one goes will be a question of how the data turn out. Indeed, a great deal of the empirical interest of the modularity thesis lies in the fact that the experimental predictions it makes tend to be diametrically opposed to the ones that New Look approaches license. But experiments to one side, there are some prima facie reasons for doubting that the computations that input systems perform could have anything like unlimited access to high-level expectations or beliefs. These considerations suggest that even if there are *some* perceptual mechanisms whose operations are extensively subject to feedback, there must be others that compute the structure of a percept largely, perhaps solely, in isolation from background information.

For one thing, there is the widely noted persistence of many perceptual illusions (e.g., the Ames room, the phi phenomenon, the Muller-Lyre illusion in vision; the phoneme restoration and click displacement effects in speech) even in defiance of the subject's explicit knowledge that the percept is illusory. The very same subject who can tell you that the Muller-Lyre arrows are identical in length, who indeed has seen them measured, still finds one looking longer than the other. In such cases it is hard to see an alternative to the view that at least *some* of the background information at the subject's disposal is inaccessible to at least some of his perceptual mechanisms.

An old psychological puzzle provides a further example of this kind. When you move your head, or your eyes, the flow of images across the retina may be identical to what it would be were the head and eyes to remain stationary while the scene moves. So: why don't we experience apparent motion when we move our eyes? Most psychologists now accept one or other version of the "corollary discharge" answer to this problem. According to this story, the neural centers which initiate head and eye motions communicate with the input analyzer in charge of interpreting visual stimulations (See Bizzi, 1968). Because the latter system knows what the former is up to, it is able to discount alterations in the retinal flow that are due to the motions of the receptive organs.

Well, the point of interest for us is that this visual-motor system is informationally encapsulated. Witness the fact that, if you (gently) push your eyeball with your finger (as opposed to moving it in the usual way: by an exercise of the will), you *do* get apparent motion. Consider the moral: when you voluntarily move your eyeball with your finger, you certainly are possessed of the information that it's your eye (and not the visual scene) that is moving. This knowledge is absolutely explicit; if I ask you, you can *say* what's going on. But this explicit information, available to you for (e.g.) report, is *not* available to the analyzer in charge of the perceptual integration of your retinal stimulations. That system has access to corollary discharges from the motor center *and to no other information that you possess.* Modularity with a vengeance.

We've been surveying first blush considerations which suggest that at least some input analyzers are encapsulated with respect to at least some sorts of feedback. The next of these is a point of principle: feedback works only to the extent that the information which perception supplies is redundant; and it *is* possible to perceptually analyze arbitrarily unredundant stimulus arrays. This point is spectacularly obvious in the case of language. If I write "I keep a giraffe in my pocket," you are able to understand me despite the fact that, on even the most inflationary construal of the notion of context, there is nothing in the context of the inscription that would have enabled you to predict either its form or its content. In short, feedback is effective only to the extent that, *prior* to the analysis of the stimulus, the perceiver knows quite a lot about what the stimulus is going to be like. Whereas, the *point* of perception is, surely, that it lets us find out how the world is even when the world is some way that we *don't* expect it to be. The teleology of perceptual capacities presupposes a considerably-less-than-omniscient-organism; they'd be no use to God. If you already *know* how things are, why look to *see* how things are?[21]

So: The perceptual analysis of *un*anticipated stimulus layouts (in language and elsewhere) is possible only to the extent that (a) the output of the transducer is insensitive to the beliefs/expectations of the organism; and (b) the input analyzers are adequate to compute a representation of the stimulus from the information that the transducers supply. This is to say that the perception of novelty depends on bottom-to-top perceptual mechanisms.

There is a variety of ways of putting this point, which is, I think, among the most important for understanding the character of the input systems. Pylyshyn (1980) speaks of the "cognitive impenetrability" of perception, meaning that the output of the perceptual systems is largely insensitive to what the perceiver presumes or desires. Pylyshyn's point is that a condition for the *reliability* of perception, at least for a fallible organism, is that it generally sees what's there, not what it wants or expects to be there. Organisms that don't do so become deceased.

Here is another terminology for framing these issues about the direction of information flow in perceptual analysis: Suppose that the organism is given the problem of determining the analysis of a stimulus at a certain level of representation—e.g., the problem of determining which sequence of words a given utterance encodes. Since, in the general case, transducer outputs underdetermine perceptual analyses,[22] we can think of the solution of such problems as involving processes of nondemonstrative inference. In particular, we can think of each input system as a computational mechanism which projects and confirms a certain class of hyputheses on the basis of a certain body of data. In the present example, the available hypotheses are the word sequences that can be constructed from entries in the subject's mental lexicon, and the perceptual problem is to determine which of these sequences provides the right analysis of the currently impinging utterance token. The mechanism which solves the problem is, in effect, the realization of a *confirmation function*: it's a mapping which associates with each pair of a lexical hypothesis and some acoustic datum a value which expresses the degree of confirmation that the latter bestows upon the former. (And similarly, mutatis mutandis, for the nondemonstrative inferences that the other input analyzers effect.) I emphasize that construing the situation this way involves no commitment to a detailed theory of the operation of perceptual systems. *Any* nondemonstrative inference can be viewed as the projection and confirmation of a hypothesis, and I take it that perceptual inferences must in general be nondemonstrative, since their underdetermination by sensory data is not in serious dispute.

Looked at this way, the claim that input systems are informationally encapsulated is equivalent to the claim that the data that can bear on the confirmation of perceptual hypotheses includes, in the general case, considerably less than the organism may know. That is, the confirmation function for input systems does not have access to all of the information that the organism internally represents; there are restrictions upon the allocation of internally represented information to input processes.

Talking about the direction of information flow in psychological processes and talking about restrictions upon the allocation of information to such processes are thus two ways of talking about the same thing. If, for example, we say that the flow of information in language comprehension runs directly from the determination of the phonetic structure of an utterance to the determination of its lexical content, then we are saying that only phonetic information is available to whatever mechanism decides the level of confirmation of perceptual hypotheses about lexical structure. On that account, such mechanisms are encapsulated with respect to *non*phonetic information; they have no access to such information; not even if it is *internally represented, accessible to other cognitive processes* (i.e., to cognitive processes other than the assignment of lexical analyses to phone sequences) and *germane* in the sense that if it *were* brought to bear in lexical analysis, it would affect the confirmation levels of perceptual hypotheses about lexical structure.

I put the issue of informational encapsulation in terms of constraints on the data available for hypothesis confirmation because doing so will help us later, when we come to compare input systems with central cognitive processes. Suffice it to say, for the moment, that this formulation suggests another possible reason why input systems are so fast. We remarked above that the computations that input systems perform are mandatory, and that their being so saves time that would otherwise have to be used in executive decision-making. We now add that input systems are bull-headed and that this, too, makes for speed. The point is this: to the extent that input systems are informationally encapsulated, of all the information that might *in principle* bear upon a problem of perceptual analysis only a portion (perhaps only quite a small and stereotyped portion) is actually admitted for consideration. This is to say that speed is purchased for input systems by permitting them to ignore lots of the facts. Ignoring the facts is not, of course, a good recipe for problem-solving in the general case. But then, as we have seen, input systems don't function in the *general* case. Rather, they function to provide very special kinds of representations of very specialized inputs (to pair transduced representations with formulas in the domains of central processes). What operates in the general case, and what is sensitive, at least in principle, to *everything* that the organism knows, are the central processes themselves. Of which more later.

I should add that these reflections upon the value of bull-headedness do not, as one might suppose, entirely depend upon assumptions about the speed of memory search. Consider an example. Ogden Nash once offered the following splendidly sane advice: "If you're called by a panther/don't anther." Roughly, we want the perceptual identification of panthers to be very fast and to err, if at all, only on the side of false positives. If there is a body of information that must be deployed in such perceptual identifications, then we would prefer not to have to recover that information from a large memory, assuming that the speed of access varies inversely with the amount of information that the memory contains. This is a way of saying that we do not, on that assumption, want to have to access panther-identification information from the (presumably *very* large) central storage in which representations of background-information-at-large are generally supposed to live. Which is in turn to say that we don't want the input analyzer that mediates panther identification to communicate with the central store on the assumption that large memories are searched slowly.

Suppose, however, that random access to a memory is *in*sensitive to its size. *Even so* panther-identification (and, mutatis mutandis, other processes of input analysis) had better be insensitive to much of what one knows. Suppose that we can get at *everything* we know about panthers *very fast*. We still have the problem of deciding, for each such piece of information retrieved from memory, *how much inductive confirmation it bestows upon the hypothesis that the presently observed black-splotch-in-the-visual-field is a panther.* The point is that in the rush and scramble of panther identification, there are many things I know about panthers whose bearing on the likely pantherhood of the present stimulus *I do not wish to have to consider.* As, for example, that my grandmother abhors panthers; that every panther bears some distant relation to my Siamese cat Jerrold J.; that there are no panthers on Mars; that there is an Ogden Nash poem about panthers . . . etc. Nor is this all; for, in fact, the property of being 'about panthers' is not one that can be surefootedly relied upon. Given enough context, practically everything I know can be construed as panther related; and, *I do not want to have to consider everything I know* in the course of perceptual panther identification. In short, the point of the informational encapsulation of input processes is not—or not solely—to reduce the memory space that must be searched to find information that is perceptually relevant. The primary point is to so restrict the number of *confirmation relations* that need to be estimated as to make perceptual identifications fast. (I am indebted to Scott Fahlman for raising questions that provoked the last two paragraphs.)[23]

The informational encapsulation of the input systems is, or so I shall argue, the essence of their modularity. It's also the essence of the analogy between the input systems and reflexes; reflexes are informationally encapsulated with bells on.

Suppose that you and I have known each other for many a long year (we were boys together, say) and you have come fully to appreciate the excellence of my character. In particular, you have come to know perfectly well that under no conceivable circumstances would I stick my finger in your eye. Suppose that this belief of yours is both explicit and deeply felt. You would, in fact, go to the wall for it. Still, if I jab my finger near enough to your eyes, and fast enough, you'll blink. To say, as we did above, that the blink reflex is mandatory is to say, inter alia, that it has no access to what you know about my character or, for that matter, to any other of your beliefs, utilities and expectations. For this reason the blink reflex is often produced when sober reflection would show it to be uncalled for; like panther-spotting, it is prepared to trade false positives for speed.

That is what it is like for a psychological system to be informationally encapsulated. If you now imagine a system that is encapsulated in the way that reflexes are, but also computational in a way that reflexes are not, you will have some idea of what I'm proposing that input systems are like.

It is worth emphasizing that being modular in this sense is not quite the same thing as being autonomous in the sense that Gall had in mind. For Gall, if I read him right, the claim that the vertical faculties are autonomous was practically equivalent to the claim that there are no horizontal faculties for them to share. Musical aptitude, for example, is autonomous *in that* judging musical ideas shares no cognitive mechanisms with judging mathematical ideas; remembering music shares no cognitive mechanisms with remembering faces; perceiving music shares no cognitive mechanisms with perceiving speech; and so forth.

Now, it is unclear to what extent the input systems *are* autonomous in *that* sense. We do know, for example, that there are systematic relations between the amount of computational strain that decoding a sentence places on the language handling systems and the subject's ability to perform simultaneous nonlinguistic tasks quickly and accurately. 'Phoneme monitor' (Foss, 1970) techniques, and others, can be used to measure such interactions, and the results suggest a picture that is now widely accepted among cognitive psychologists: Mental processes often compete for access to resources variously characterized as attention, short-term memory, or work space; and the result of allocating such resources to one of the competing processes is a decrement in the performance of the others. How general this sort of interaction is is unclear in the present state of the art (for contrary cases, suggesting isolated work spaces for visual imagery on the one hand and verbal recall on the other, see Brooks, 1968). In any event, where such competition does obtain, it is a counterexample to autonomy in what I am taking to be Gall's understanding of that notion.[24]

On the other hand, we can think of autonomy in a rather different way from Gall's—viz., in terms of informational encapsulation. So, instead of asking what access language processes (e.g.) have to *computational resources* that other systems also share, we can ask what access they have to the *information* that is available to other systems. If we do look at things this way, then the question "how much autonomy?" is the same question as "how much constraint on information flow?" In a nutshell: one way that a system can be autonomous is by being encapsulated, by not having access to facts that other systems know about. I am claiming that, whether or not the input systems are autonomous in Gall's sense, they are, to an interesting degree, autonomous in this informational sense.

However, I have not yet given any arguments (except some impressionistic ones) to show that the input systems actually are informationally encapsulated. In fact, I propose to do something considerably more modest: I want to suggest some caveats that ought to be, but frequently aren't, observed in interpreting the sorts of data that have usually been alleged in support of the contrary view. I think that many of the considerations that have seemed to suggest that input processes are cognitively penetrable—that they are importantly affected by the subject's belief about context, or his background information, or his utilities—are, in fact, equivocal or downright misleading. I shall therefore propose several ground rules for evaluating claims about the cognitive penetrability of input systems; and I'll suggest that, when these rules are enforced, the evidence for 'New Look' approaches to perception begins to seem not impressive. My impulse in all this is precisely analogous to what Marr and Pogio say motivates their work on vision: ". . . to examine ways of squeezing the last ounce of information from an image before taking recourse to the descending influence of high-level interpretation on early processing" (1977, pp. 475–476).

(a) Nobody doubts that the information that input systems provide must somehow be reconciled with the subject's background knowledge. We sometimes know that the world can't really be the way that it looks, and such cases may legitimately be described as the correction of input analyses by top-down information flow. (This, ultimately, is the reason for refusing to identify input analysis with perception. The point of perception is the fixation of belief, and the fixation of belief is a *conservative* process—one that is sensitive, in a variety of ways, to what the perceiver already knows. Input analysis may be informationally encapsulated, but perception surely is not.) However, to demonstrate *that* sort of interaction between input analyses and background knowledge is not, in and of itelf, tantamount to demonstrating the cognitive penetrability of the former; you need also to show that the locus of the top-down effect is *internal* to the input system. That is, you need to show that the information fed back interacts with interlevels of input-processing and not merely with the final results of such processing. The penetrability of a system is, by definition, its susceptibility to top-down effects at stages *prior* to its production of output.

I stress this point because it seems quite possible that input systems specify only relatively shallow levels of representation (see the next section). For example, it is quite possible that the perceptual representation delivered for a token sentence specifies little more than the type to which the token belongs (and hence does *not* specify such information as the speech act potential of the token, still less the speech act performed by the tokening). If this is so, then data showing effects of the hearer's background information on, e.g., his estimates of the speaker's communicative intentions would *not* constitute evidence for the cognitive penetration of the presumptive language-comprehension module; by hypothesis, the computations involved in making such estimates would not be among those that the language-comprehension module *per se* performs. Similarly, mutatis mutandis, in the case of vision. There is a great deal of evidence for context effects upon certain aspects of visual object recognition. But such evidence counts for nothing in the present discussion unless there is independent reason to believe that these aspects of object recognition are part of visual input analysis. Perhaps the input system for vision specifies the stimulus only in terms of "primal sketches" (for whose cognitive impenetrability there is, by the way, some nontrivial evidence. See Marr and Nishihara (1978).) The problem of assessing the degree of informational encapsulation of input systems is thus not independent of the problem of determining how such systems are individuated and what sorts of representations constitute their outputs. I shall return to the latter issue presently; for the moment, I'm just issuing caveats.

(b) Evidence for the cognitive penetrability of some computational mechanism that does what input systems do is not, in and of itself, evidence for the cognitive penetrability of input systems.

To see what is at issue here, consider some of the kinds of findings that have been taken as decisively exhibiting the effects of background expectations upon language perception. A well known way of estimating such expectations is the use of the so-called Cloze procedure. Roughly, S is presented with the first *n* words of a sentence and is asked to complete the fragment. Favored completions (as, for example, "salt" in the case of the fragment "I have the pepper, but would you please pass the ----") are said to be "high Cloze" and are assumed to indicate what the subject would expect a speaker to say next if he had just uttered a token of the fragment. An obvious generalization allows the estimation of the Cloze value at each point in a sentence, thereby permitting experiments in which the average Cloze value of the stimulus sentences is a manipulated variable.

It is quite easy to show that relative Cloze value affects S's performance on a number of experimental tasks, and it is reasonable to infer from such demonstrations that whatever mechanisms mediate the performance of these tasks must have access to S's expectations about what speakers are likely to say, hence not just to the 'stimulus' (e.g., acoustic) properties of the linguistic token under analysis. (For an early review of the literature on redundancy effects in sentence processing, see Miller and Isard, 1963.) So, for example, it can be shown that the accuracy of S's perception of sentences heard under masking noise is intimately related to the average Cloze value of the sentences: high Cloze sentences can be understood under conditions of greater distortion than the perception of low Cloze sentences tolerates. (Similarly, high Cloze sentences are, in general, more easily remembered than low Cloze sentences; recognition thresholds for words that are high Cloze in a context are lower than those for words that are low Cloze in that context; and so forth.)

The trouble with such demonstrations, however, is that although they show that there exist *some* language-handling processes that have access to the hearer's expectations about what is likely to be said, they do *not* show that the input systems enjoy such access. For example, it might be argued that, in situations where the stimulus is acoustically degraded, the subject is, in effect, encouraged to guess the identity of the material that he can't hear. (Similarly, mutatis mutandis, in memory experiments where a reasonable strategy for the subject is to guess at such of the material as he can't recall.) Not surprisingly, in such circumstances, the subject's background information comes into play with measurable effect. The question, however, is whether the psychological mechanisms deployed in the slow, relatively painful, highly attentional process of reconstructing noisy or otherwise degraded linguistic stimuli are the same mechanisms which mediate the automatic and fluent processes of normal speech perception.

That this question is not merely frivolous is manifested by results such as those of Fishler and Bloom (1980). Using a task in which sentences are presented in clear, they found only a marginal effect of high Cloze on the recognition of test words, and such effects vanished entirely when the stimuli were presented at high rates. (High presentation rates presumably discourage guessing; guessing takes time.) By contrast, words that are 'semantically anomalous' in context showed considerable inhibition in comparison with neutral controls. This last finding is of interest because it suggests that at least some of the effects of sentence context in speech recognition must be, as psychologists sometimes put it, 'post-perceptual'. In our terminology, these processes must operate *after* the input system has provided a (tentative) analysis of the lexical content of the stimulus. The point is that even if the facilitation of redundant items is mediated by predictive, expectation-driven mechanisms, the inhibition of contextually anomalous items cannot be. It is arguable that, in the course of speech perception, one is forever making such predictions as that 'pepper' will occur in 'salt and ----; but surely one can't also be forever predicting that 'dog', 'tomorrow', and all the other anomalous expressions will *not* occur there.[25] The moral is: some processes which eventuate in perceptual identifications are, doubtless, cognitively penetrated. But this is compatible with the informational encapsulation of the input systems themselves. Some traditional enthusiasm for context-driven perceptual models may have been prompted by confusion on this point.

(c) The claim that input systems are informationally encapsulated must be very carefully distinguished from the claim that there is top-down information flow *within* these systems. These issues are very often run together, with consequent exaggeration of the well-groundedness of the case against encapsulation.

Consider, once again, the phoneme restoration effect. Setting aside the general caution that experiments with distorted stimuli provide dubious grounds for inferences about speech perception in clear, phoneme restoration provides considerable prima facie evidence that phone identification has access to what the subject knows about the lexical inventory of his language. If this interpretation is correct, then phoneme restoration illustrates top-down information flow in speech perception. It does *not*, however, illustrate the cognitive penetrability of the language input system. To show that that system is penetrable (hence informationally unencapsulated), you would have to show that its processes have access to information that is not specified at any of the levels of representation that the language input system computes; for example, that it has generalized access to what the hearer knows about the probable beliefs and intentions of his interlocutors. If, by contrast, the 'background information' deployed in phoneme restoration is simply the hearer's knowledge of the words in his language, then that counts as top-down flow within the language module; on any remotely plausible account, the knowledge of a language includes knowledge of its lexicon.

The most recent work in phoneme restoration makes this point with considerable force. Samuel (1981) has shown that both information about the lexical inventory and 'semantic' information supplied by sentential context affect the magnitude of the phoneme restoration effect. Specifically, you get more restoration in words than in (phonologically possible) nonwords, and you get more restoration when a word is predictable in sentence context than when the context is neutral. This looks like the penetration of phone recognition by both lexical and 'background' information, but the appearance is misleading. In fact, Samuel's data suggest that, of the two effects, *only the former* is strictly perceptual, the latter operating in consequence of a response bias to report predictable words as intact. (Detection theoretically: the word/nonword difference affects d′, whereas the neutral context/predictive context difference affects β.) As Samuel points out, the amount of restoration is inversely proportional to S's ability to distinguish the stimulus word with a phone missing from an undistorted token of the same type; and, on Samuel's data, this discrimination is actually *better* for items that are highly predictable in context than for items that aren't. Another case, in short, where what had been taken to be an example of context-driven prediction in perception is, in fact, an effect of the biasing of post-perceptual decision processes.

The importance of distinguishing cognitive penetration from intramodular effects can be seen in many other cases where predictive analysis in perception is demonstrable. It is, for example, probable (though harder to show than one might have supposed) that top-down processes are involved in the identification of the surface constituent structure of sentences (see Wright, 1982). For example, it appears that the identification of nouns is selectively facilitated in contexts like T A ------, the identification of verbs is selectively facilitated in contexts like T N ------, and so forth. Such facilitation indicates that the procedures for assigning lexical items to form classes have access to information about the general conditions upon the well-formedness of constituent structure trees.

Now, it is a question of considerable theoretical interest whether, and to what extent, predictive analysis plays a role in parsing; but this issue must be sharply distinguished from the question whether the parser is informationally encapsulated. Counterexamples to encapsulation must exhibit the sensitivity of the parser to information that is not specified internal to the language-recognition module, and constraints on syntactic well-formedness are paradigms of information that does *not* satisfy this condition. The issue is currently a topic of intensive experimental and theoretical inquiry; but as things stand I know of *no* convincing evidence that syntactic parsing is ever guided by the subject's appreciation of semantic context or of 'real world' background. Perhaps this is not surprising; there are, in general, so many syntactically different ways of saying the same thing that even if context allowed you to estimate the *content* of what is about to be said, that information wouldn't much increase your ability to predict its *form*.[26]

These questions about where the interacting information comes from (whether it comes from inside or outside the input system) take on a special salience in light of the following consideration: it is possible to imagine ways in which mechanisms *internal* to a module might contrive to, as it were, mimic effects of cognitive penetration. The operation of such mechanisms might thus invite overestimations of the extent to which the module has access to the organism's general informational resources. To see how this might occur, let's return to the question of contextual facilitation of word recognition; traditionally a parade case for New Look theorizing, but increasingly an area in which the data are coming to seem equivocal.

Here are the bare bones of an ingenious experiment of David Swinney's (1979; for further, quite similar, results, see Tannenhaus, Leirnau, and Seidenberg, 1979). The subject listens to a stimulus sentence along the lines of "Because he was afraid of electronic surveillance, the spy carefully searched the room for bugs." Now, we know from previous research that the response latencies for 'bugs' (say, in a word/nonword decision task) will be faster in this context, where it is relatively predictable, than in a neutral context where it is acceptable but relatively low Cloze. This seems to be—and is traditionally taken to be—the sort of result which demonstrates how expectations based upon an intelligent appreciation of sentential context can guide lexical access; the subject predicts 'bugs' before he hears the word. His responses are correspondingly accelerated whenever his prediction proves true. Hence, cognitive penetration of lexical access.

You can, or so it seems, gild this lily. Suppose that, instead of measuring reaction time for word/nonword decisions on 'bugs', you simultaneously present (flashed on a screen that the subject can see) a different word belonging to the same (as one used to say) 'semantic field' (e.g., 'microphones'). If the top-down story is right in supposing that the subject is using semantic/background information to predict lexical content, then 'microphones' is as good a prediction in context as 'bugs' is, so you might expect that 'microphones', too, will exhibit facilitation as compared with a neutral context. And so it proves to do. Cognitive penetration of lexical access with bells on, or so it would appear.

But the appearance is misleading. For Swinney's data show that if you test with 'insects' instead of 'microphones', you get the same result: facilitation as compared with a neutral context. Consider what this means. 'Bugs' has two paraphrases: 'microphones' and 'insects'. But though only one of these is contextually relevant, *both are contextually facilitated.* This looks a lot less like the intelligent use of contextual/background information to guide lexical access. What it looks like instead is some sort of associative relation among lexical forms (between, say, 'spy' and 'bug'); a relation pitched at a level of representation sufficiently superficial to be *in*sensitive to the semantic content of the items involved. This possibility is important for the following reason: If facilitation is mediated by merely interlexical relations (and not by the interaction of background information with the semantic content of the item and its context), then the information that is exploited to produce the facilitation can be represented *in the lexicon;* hence *internal to the language recognition module.* And if that is right, then contextual facilitation of lexical access is *not* an argument for the cognitive penetration of the module. It makes a difference, as I remarked above, where the penetrating information comes from.

Let's follow this just a little further. Suppose the mental lexicon is a sort of connected graph, with lexical items at the nodes and with paths from each item to several others. We can think of accessing an item in the lexicon as, in effect, exciting the corresponding node; and we can assume that one of the consequences of accessing a node is that excitation spreads along the pathways that lead from it. Assume, finally, that when excitation spreads through a portion of the lexical network, response thresholds for the excited nodes are correspondingly lowered. Accessing a given lexical item will thus decrease the response times for items to which it is connected. (This picture is familiar from the work of, among others, Morton, 1969, and Collins and Loftus, 1975; for relevant experimental evidence, see Meyer and Schvaneveldt, 1971.)

The point of the model-building is to suggest how mechanisms internal to the language processor could mimic the effects that cognitive penetration would produce if the latter indeed occurred. In the present example, what mimics the background knowledge that (roughly) spies have to do with bugs is the existence of a connection betweeen the node assigned to the word 'spy' and the node assigned to the word 'bug'. Facilitation of 'bug' in spy contexts is affected by the excitation of such intralexical connections.

Why should these intralexical connections exist? Surely not just in order to lead psychologists to overestimate the cognitive penetrability of language-processing. In fact, if one works the other way 'round and assumes that the input systems are encapsulated, one might think of the mimicry of penetration as a way that the input processors contrive to make the best of their informational isolation. Presumably, what encapsulation buys is speed; and, as we remarked above, it buys speed at the price of unintelligence. It would, one supposes, take a lot of time to make reliable decisions about whether there is the kind of relation between spies and bugs that makes it on balance likely that the current token of 'spy' will be followed by a token of 'bug'. But that is precisely the kind of decision that the subject would have to make if the contextual facilitation of lexical access were indeed an effect of background knowledge interacting with the semantic content of the context. The present suggestion is that no such intelligent evaluation of the options takes place; there is merely a brute facilitation of the recognition of 'bug' consequent upon the recognition of 'spy'. The condition of this brute facilitation buying anything is that it should be possible, with reasonable accuracy, to mimic what one knows about connectedness *in the world* by establishing corresponding connections among entries in the mental lexicon. In effect, the strategy is to use the structure of interlexical connections to mimic the structure of knowledge. The mimicry won't be precise (a route from 'spy' to 'insect' will be generated as a by-product of the route from 'spy' to 'bug'). But there's no reason to doubt that it may produce savings over all.

Since we are indulging speculations, we might as well indulge this one: It is a standing mystery in psychology why there should be interlexical associations at all; why subjects should exhibit a reliable and robust disposition to associate 'salt' with 'pepper', 'cat' with 'dog', 'mother' with 'father', and so forth. In the heyday of associationism, of course, such facts seemed quite *un*mysterious; they were, indeed, the stuff of which the mental life was supposed to be made. On one account the utterance of a sentence was taken to be a chained response, and associations among lexical items were what held the links together. According to still earlier tradition, the postulation of associative connections between Ideas was to be the mechanism for reconstructing the notion of degree of belief. None of this seems plausible now, however. Belief is a matter (not of association but) of *judgment*; sentence production is a matter (not of association but) of *planning*. So, what on earth are associations *for*?

The present suggestion is that associations are the means whereby stupid processing systems manage to behave as though they were smart ones. In particular, interlexical associations are the means whereby the language processor is enabled to act as though it knows that spies have to do with bugs (whereas, in fact, it knows no such thing). The idea is that, just as the tradition supposed, terms for things frequently connected in experience become themselves connected in the lexicon. Such connection is *not* knowledge; it is not even judgment. It is simply the mechanism of the contextual adjustment of response thresholds. Or, to put the matter somewhat metaphysically, the formation of interlexical connections buys the synchronic encapsulation of the language processor at the price of its cognitive penetrability *across time*. The information one has about how things are related in the world is inaccessible to modulate lexical access; that is what the encapsulation of the language processor implies. But one's experience of the relations of things in the world does affect the structure of the lexical network—viz., by instituting connections among lexical nodes. If the present line of speculation is correct, these connections have a real, if modest, role to play in the facilitation of the perceptual analysis of speech. The traditional, fundamental, and decisive objection to association is that it is too stupid a relation to form the basis of a mental life. But stupidity, when not indulged in to excess, is a virtue in fast, peripheral processes; which is exactly what I have been supposing input processes to be.

I am not quite claiming that all the putative effects of information about background (context, etc.) on sentence recognition are artifacts of connections in the lexical network (though, as a matter of fact, such experimental attempts as I've seen to demonstrate a residual effect of context after interlexical/associative factors are controlled for strike me as not persuasive). I am claiming only that the possibility of such artifacts contaminates quite a lot of the evidence that is standardly alleged. The undoubted fact that "semantically" coherent text is relatively easy to process does not, in and of itself, demonstrate that the input system for language has access to what the organism knows about how the world coheres. Such experimental evidence as supported early enthusiasms for massively top-down perceptual models was, I think, sexy but inconclusive; and the possibility of a modular treatment of input processes provides motivation for its reconsideration. The situation would seem to be paradigmatically Kuhnian: the data look different to a jaundiced eye.

Consider the provenance of New Look theorizing. Cognitive psychologists in the '40s and '50s were faced with the proposal that perception is *literally* reflexive; for example, that the theory of perception is reducible without residue to the theory of discriminative operant response. It was natural and admirable in such circumstances to stress the 'intelligence' of perceptual integration. However, in retrospect it seems that the intelligence of perceptual integration may have been seriously misconstrued by those who were most its partisans.

In the ideal condition—one approached more frequently in the textbooks than *in rerum naturae,* to be sure—reflexes have two salient properties. They are computationally simple (the stimulus is "directly connected" to the response), and they are informationally encapsulated (see above). I'm suggesting that New Look theories failed to distinguish these properties. They thus assumed, wrongly, that the disanalogy between perceptual and reflexive processes consisted in the capacity of the former to access and exploit background information. From the point of view of the modularity thesis, this is a case of the right intuition leading to the wrong claim. Input systems *are* computationally elaborated. Their typical function is to perform inference-like operations on representations of impinging stimuli. Processes of input analysis are thus unlike reflexes in respect of the character and complexity of the operations that they perform. But this is quite compatible with reflexes and input processes being similar in respect of their informational encapsulation; in this latter respect, both of them contrast with "central processes"—problem-solving and the like—of which cognitive penetrability is perhaps the most salient feature, or so I shall argue below. To see that informational encapsulation and computational elaboration are compatible properties, it is only necessary to bear in mind that unencapsulation is the exploitation of information from *outside* a system; a computationally elaborated sytem can thus be encapsulated if it stores the information that its computations exploit. Encapsulation is a matter of foreign affairs; computational elaboration begins at home.

It may be useful to summarize this discussion of the informational encapsulation of input systems by comparing it with some recent, and very interesting, suggestions owing to the philosopher Steven Stich (1978). Stich's discussion explores the difference between belief and the epistemic relation that is alleged to hold between, for example, speaker/hearers and the grammar of their native language (the relation that Chomsky calls 'cognizing'). Stich supposes, for purposes of argument, that the empirical evidence shows that speakers *in some sense* 'know' the grammar of their native language; his goal is to say something about what that sense is.

Let us call the epistemic relation that a native speaker has to the grammar of his language *subdoxastic* belief.[27] Stich suggests that there are two respects in which subdoxastic beliefs differ from beliefs strictly so-called. In the first place, as practically everybody has emphasized, subdoxastic beliefs are *unconscious*. But, Stich adds, subdoxastic beliefs are also typically "inferentially unintegrated." The easiest way to understand what Stich means by this is to consider one of his examples.

> If a linguist believes a certain generalization to the effect that no transformation rule exhibits a certain characteristic, and if he comes to (nonsubdoxastically) believe a given transformation which violates the generalization, he may well infer that the generalization is false. But merely having the rule stored (in the way that we are assuming all speakers of the language do) does not enable the linguist to draw the inference. . . . Suppose that for some putative rule, you have come to believe that if r then Chomsky is seriously mistaken. Suppose further that, as it happens, r is in fact among the rules stored by your language processing mechanism. The belief along with the subdoxastic state will not lead to the belief that Chomsky is seriously mistaken. By contrast, if you believe (perhaps even mistakenly) that r, then the belief that Chomsky is seriously mistaken is likely to be inferred. [pp. 508–509]

Or, as Stich puts the argument at another point, "It is characteristic of beliefs that they generate further beliefs via inference. What is more, beliefs are inferentially promiscuous. Provided with a suitable set of supplementary beliefs, almost any belief can play a role in the inference to any other. . . . (However) subdoxastic states, as contrasted with beliefs, are largely inferentially isolated from the large body of inferentially integrated beliefs to which a subject has (conscious) access."

Now, as Stich clearly sees, the proposal that subdoxastic states are typically both unconscious and inferentially unintegrated raises a question—viz., *Why should these two properties co-occur?* Why should it be, to put it in my terminology, that subdoxastic states are typically encapsulated with respect to the processes which affect the inferential integration of beliefs?

Notice that there is a kind of encapsulation that follows from unconsciousness: an unconscious belief cannot play a role as a premise in the sort of reasoning that goes on in the conscious drawing of inferences. Stich is, however, urging something more interesting than this trivial truth. Stich's claim is that subdoxastic beliefs are largely inaccessible even to *un*conscious mental processes of belief fixation. If this claim is true, the question does indeed arise why it should be so.

I want to suggest, however, that the question doesn't arise because, as a matter of act, subdoxastic beliefs are not in general encapsulated; or, to put it more precisely, they are not in general encapsulated *qua* subdoxastic. Consider, as counterexamples, one's subdoxastic views about inductive and deductive warrant; for example, one's subdoxastic acquiescence in the rule of *modus ponens*. On the sort of psychological theory that Stich has in mind, subdoxastic knowledge of such principles must be accessible to practically all mental processes, since practically all inferential processes exploit them in one way or another. One's subdoxastic beliefs about validity and confirmation are thus quite unlike one's subdoxastic beliefs about the rules of grammar; though both are unconscious, the former are paradigms of promiscuous and unencapsulated mental states. So the connection between unconsciousness and encapsulation cannot be *intrinsic*.

Nevertheless, I think that Stich is onto something important. For, though much unconscious information must be widely accessible to processes of fixation of belief, it is quite true that very many of the examples of unconscious beliefs for which there is currently good empirical evidence are encapsulated. This is because most of our current cognitive science is the science of input systems, and, as we have seen, *informational encapsulation is arguably a pervasive feature of such systems*. Input systems typically do not exchange subdoxastic information with central processes or with one another.

Stich almost sees this point. He says that "subdoxastic states occur in a variety of separate, special purpose cognitive systems" (p. 508). True enough; but they must also occur in integrated, general purpose systems (in what I'm calling "central" systems), assuming that much of the fixation of belief is both unconscious